

popquiz energy, work

 This is a preview of the published version of the quiz

Started: Apr 29 at 8:04pm

Quiz Instructions



Question 1 14.5 pts

A car has a speed of 30mph. And a KE of 80,000 J

Now the car has a speed of 60mph . Its new KE is:

(simple math. if the speed is x .. then KE is x ...)



40,000J



160,000J



320,000J



80,000J



Question 2 14.5 pts

It's better to place a hand in a 400F oven than into boiling water because



the molecules in the boiling water are moving faster



the molecules in the oven are moving slower



They are more molecules in the boiling water



There is no connection between temperature and the speed of the molecules



Question 3 14.5 pts

The work done in pushing a TV set a distance of 2 m with an average force of 20 N is



2J



10J



40J



20J



Question 4 14.5 pts

The work you do when pushing a shopping cart twice as far while applying the same force is



twice as much.



four times as much.



half as much.



the same amount.



Question 5 14.5 pts

a box pushed along a path is acted upon by a friction force = 100N. The box is pushed around a circle of radius 1m. The total work done by the friction force on the box (along the path) is

!!! triky question. Friction is a non-conservative work so the length path matters



about -600J



about 100J



0



about 600J



Question 6 14.5 pts

A Dumbbell (10kg) is lifted up to a height (1m) then lowered down to the same original position. (up and down)

During this time (up and down), the work done by gravity on the dumbbell is

hint: remember the work done by conservative force (gravity) only depends on $h_f - h_i$

☐

200J

☐

-100J

☐

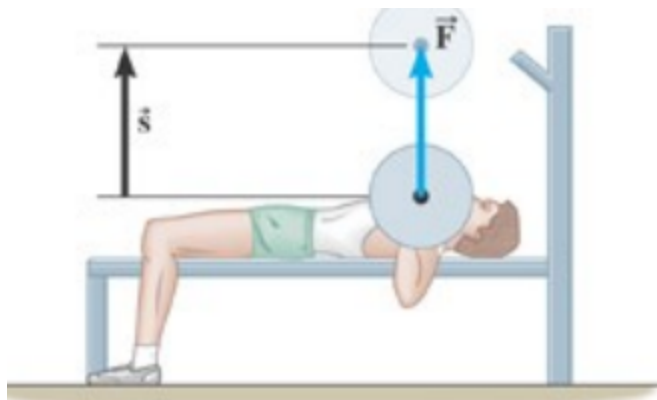
100J

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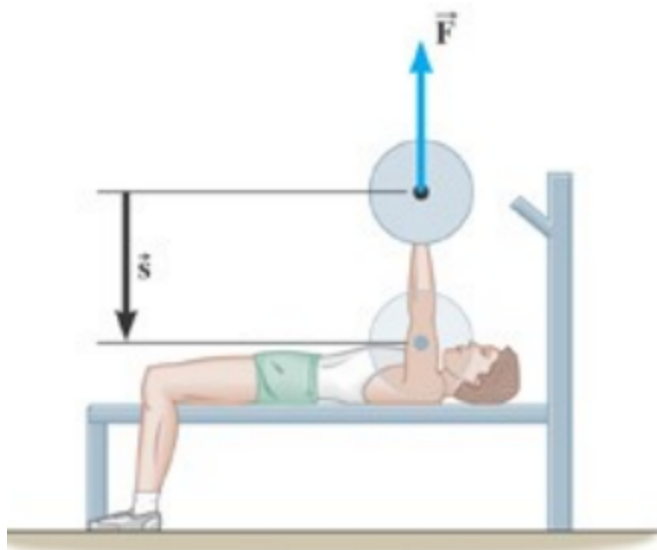
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Question 7 14.5 pts



(b)



(1) you lift the dumbbell (2) you lower the dumbbell

What is true

☐

(1) work done by gravity is 0 and (2) work done by gravity is 0

☐

(1) gravity is going negative work (2) gravity is doing 0 work

☐

for (1) and (2) gravity is doing negative work in both cases

☐

(1) gravity is going negative work (2) gravity is doing positive work

Not saved

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